

New Bounds for the Grothendieck Constant

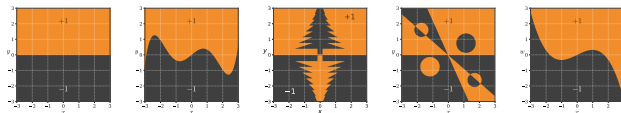
A Case Study in Long Horizon AI Research

Rahul Saha¹ Alan Li¹

with Anton Xue¹, Swarat Chaudhuri¹, Adam Klivans¹, Pravesh K. Kothari², Raghu Meka³

¹UT Austin ²Princeton ³UCLA

September 2026

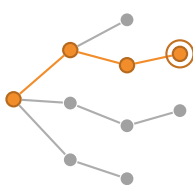


This talk



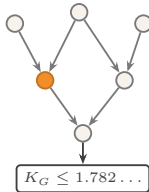
The problem

A 70 year old
unknown constant



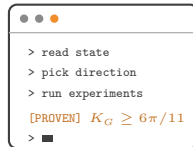
Search

LLM-guided constructions



Proof

Refining Lean Blueprint



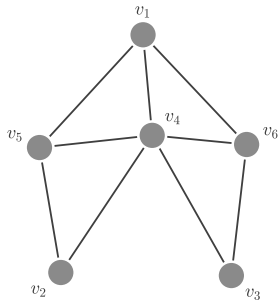
Research

Agents running
the program

Year-long AI-for-math project on single research problem.

Max-Cut

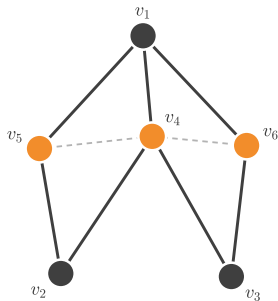
- ▶ Split the vertices of a graph into two groups so that as many edges as possible run *between* the groups
- ▶ For $S \subseteq V$, the cut edges are $\delta(S) = \{\{v_i, v_j\} \in E : v_i \in S, v_j \notin S\}$, and $\text{MaxCut}(G) = \max_S |\delta(S)|$



six vertices, nine edges

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- ▶ Here $S = \{v_4, v_5, v_6\}$ cuts 7 of the 9 edges, and nothing does better



filled vertices are in S ; dashed edges are not cut

Max-Cut is a quadratic program

- ▶ Encode a cut by signs: $x_i = +1$ if $v_i \in S$, else $x_i = -1$. An edge is cut exactly when its endpoints differ, and that indicator is a quadratic:

$$\mathbf{1}\{x_i \neq x_j\} = \frac{(x_i - x_j)^2}{4}$$

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- ▶ So the cut size is a sum of quadratics, and any such sum is $x^\top Ax$ for a symmetric A :

$$\text{MaxCut}(G) = \frac{1}{4} \max_{x \in \{\pm 1\}^n} x^\top Ax \quad (A \text{ is the graph Laplacian})$$

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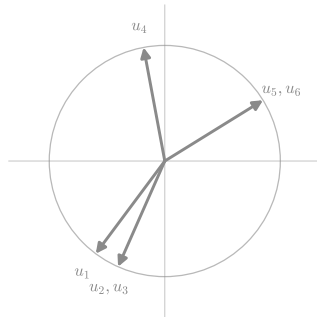
- ▶ Optimising over signs makes it **NP-hard**

Goemans–Williamson

1. **Relax.** Replace each sign x_i by a unit vector u_i , each product $x_i x_j$ by $\langle u_i, u_j \rangle$:

$$\text{SDP}_{\text{cut}}(G) = \max_{\|u_i\|=1} \frac{1}{2} \sum_{\{v_i, v_j\} \in E} (1 - \langle u_i, u_j \rangle)$$

2. **Solve.** A semidefinite program — polynomial time



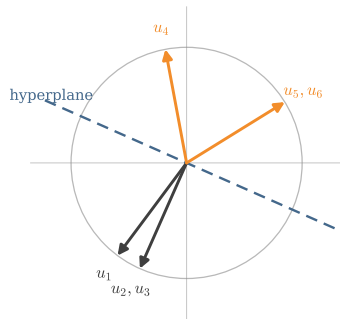
optimal unit vectors in $\mathbb{R}^2$; symmetric vertices share a vector

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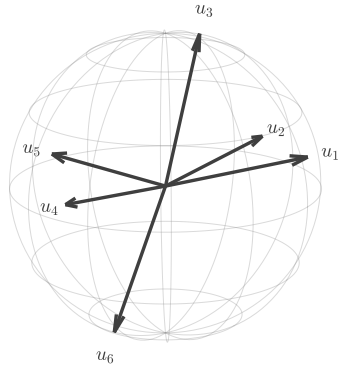
2. **Solve.** A semidefinite program — polynomial time
3. **Round.** Back to signs with a random hyperplane through the origin: one side is $+1$, the other -1



this hyperplane returns $S = \{v_4, v_5, v_6\}$

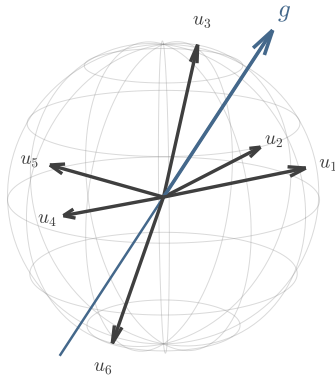
What the rounding step does

- ▶ The SDP gives us unit vectors



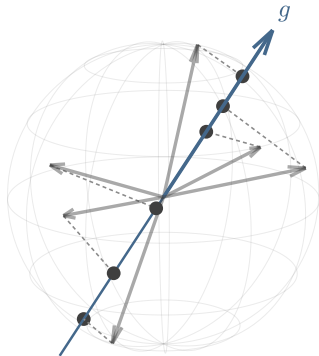
What the rounding step does

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- ▶ Pick a random direction g — a line through the origin



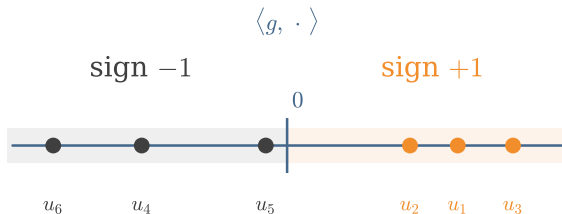
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- ▶ Project every vector onto that line



What the rounding step does

- ▶ The SDP gives us unit vectors
- ▶ Pick a random direction g — a line through the origin
- ▶ Project every vector onto that line
- ▶ Keep the sign of the number you get



The price of rounding

Goemans–Williamson (1995) The rounded cut satisfies

$$\mathbb{E} |\delta(S)| \geq \alpha_{GW} \cdot \text{MaxCut}(G), \quad \alpha_{GW} = 0.87856 \dots$$

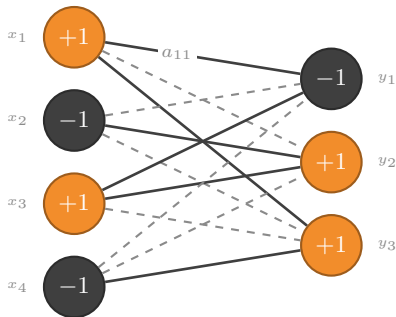
- ▶ The relaxation is solved exactly — the loss is all in the rounding
- ▶ Khot–Kindler–Mossel–O’Donnell (2007): under the Unique Games Conjecture **Theorem (?)**,
no polynomial-time algorithm does better

6 Oct 2026 — one day before this talk — an AI-written proof was posted. Lean-verified for 2-to-1; the full conjecture is still being checked.

The general problem

- ▶ Max-Cut with two *separately* chosen sign vectors and an arbitrary matrix $A \in \mathbb{R}^{m \times n}$:

$$\text{OPT}(A) = \max_{\substack{x \in \{\pm 1\}^m \\ y \in \{\pm 1\}^n}} x^\top A y = \max_{x, y} \sum_{i, j} a_{ij} x_i y_j$$

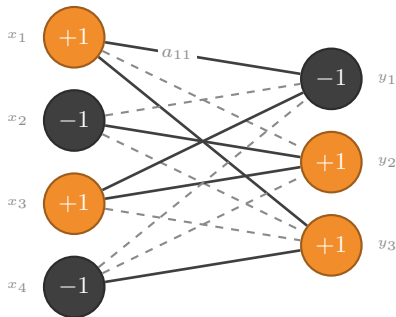


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- ▶ Captures Max-Cut, the cut norm, the bipartite Ising model, XOR games

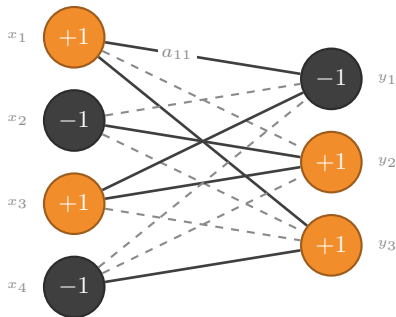


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- ▶ Captures Max-Cut, the cut norm, the bipartite Ising model, XOR games
- ▶ Still NP-hard



The same relaxation, and its price

- ▶ Replace the signs by unit vectors again, in *any* dimension:

$$\text{SDP}(A) = \max_{\|u_i\|=\|v_j\|=1} \sum_{i,j} a_{ij} \langle u_i, v_j \rangle \geq \text{OPT}(A)$$

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$$K_G = \sup_A \frac{\text{SDP}(A)}{\text{OPT}(A)}$$

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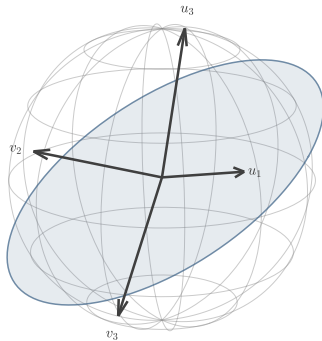
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■ 70 years on, the value of K_G is still unknown

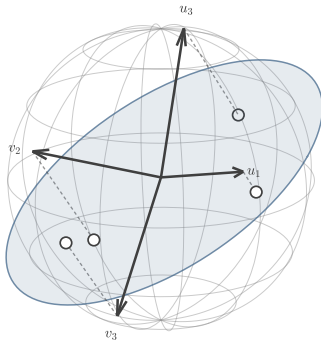
Turns out hyperplane rounding is suboptimal; what if we generalize?

- Project onto a random plane instead of a random line



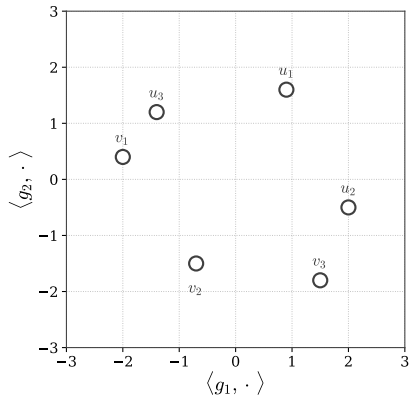
Turns out hyperplane rounding is suboptimal; what if we generalize?

- ▶ Project onto a random **plane** instead of a random line
- ▶ Every vector drops onto that plane
- ▶ It now lands at a *point*, not a number



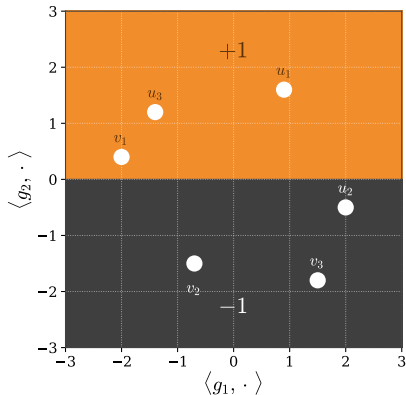
Now colour the plane

- ▶ Look at that plane face-on — one point per vector



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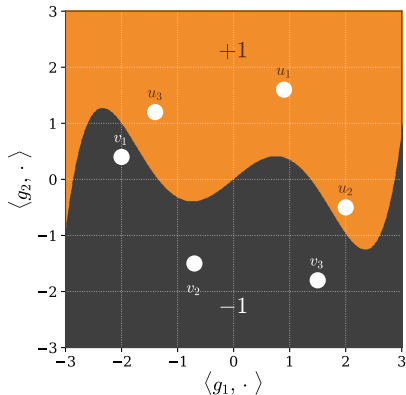
- ▶ Look at that plane face-on — one point per vector
- ▶ Colour it; the colour is the vector's sign
- ▶ A half-plane through 0 is exactly hyperplane rounding



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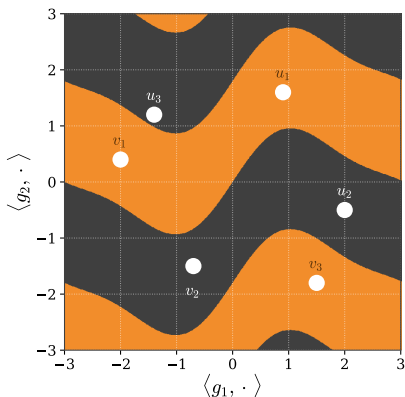
- ▶ Look at that plane face-on — one point per vector
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- ▶ Bend the boundary and some vectors switch sides
 (u_2, v_1)
- ▶ Any colouring at all gives a way to round

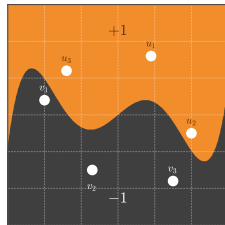


The colouring is a partition $f : \mathbb{R}^k \rightarrow \{\pm 1\}$ — and k need not be 2

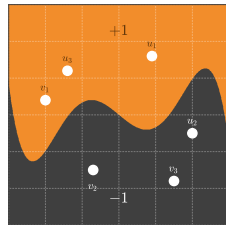
Krivine schemes

- Krivine (1977): preprocess the vectors, *then* round

$$K_G \leq \pi / (2 \log(1 + \sqrt{2})) = 1.78221 \dots$$



f (rounds x)



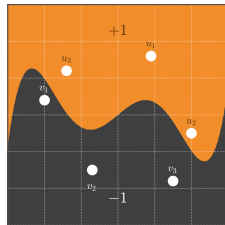
g (rounds y)

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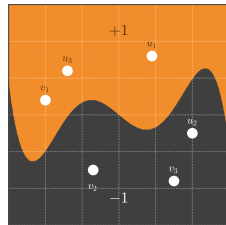
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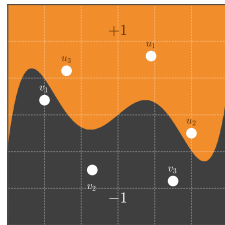
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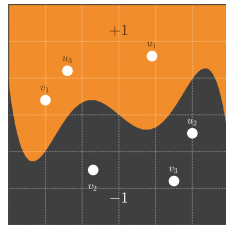
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- ▶ A scheme is a **pair** of odd partitions
 $f, g : \mathbb{R}^k \rightarrow \{\pm 1\}$



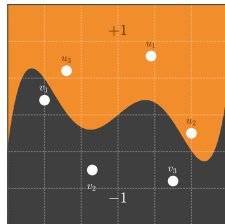
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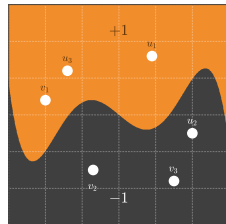
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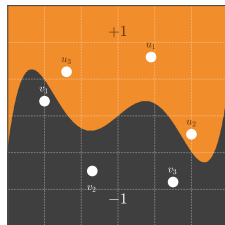
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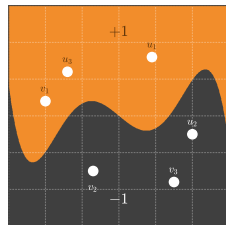
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f (rounds x)



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Improving the bound on K_G = searching for better (f, g)

Seventy years of progress

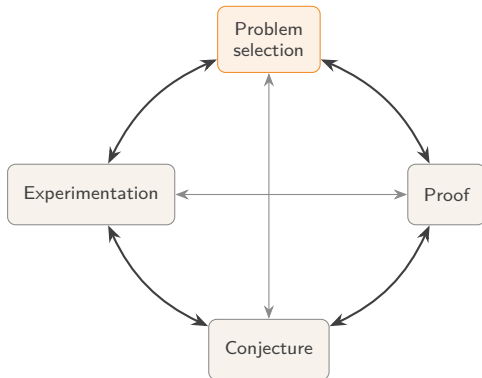
		lower	upper
1953	Grothendieck	$\pi/2 = 1.5707\dots$	$\sinh(\pi/2) = 2.3012\dots$
1977	Krivine		$\frac{\pi}{2 \log(1+\sqrt{2})} = 1.78221\dots$
1984, 91	Davie, Reeds	1.67696...	
2011	Braverman et al.		strictly below Krivine, no number
2014	Naor–Regev	Krivine schemes are optimal (a framework, not a number)	

$$1.67696 \leq K_G \leq 1.78221$$

$$K_G = 1.\textcolor{brown}{?} \dots$$

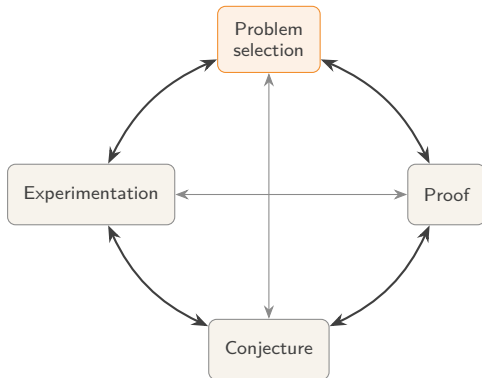
three decades stuck — not even the tenths digit

Mathematical research as an interactive process



- ▶ Different components with different goals

Mathematical research as an interactive process



- ▶ Different components with different goals
- ▶ AI-for-mathematics: moving from dense feedback to sparse feedback

May 2026: A Breakthrough

- ▶ Proved the 10^{-5} improvement from Act 1!

May 2026: A Breakthrough

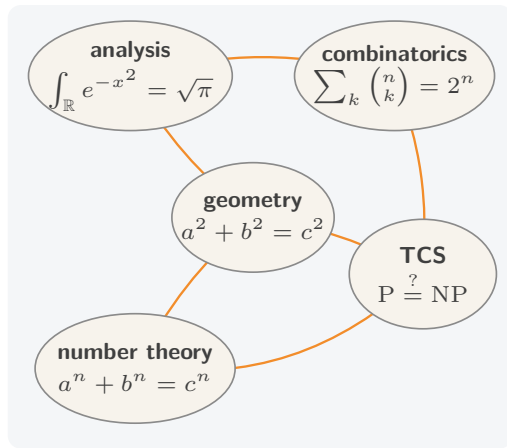
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May 2026: A Breakthrough

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- ▶ In the process, answers a 15-year-old question
- ▶ Realized the capability of LLMs to come up with new mathematics

Potential of LLMs

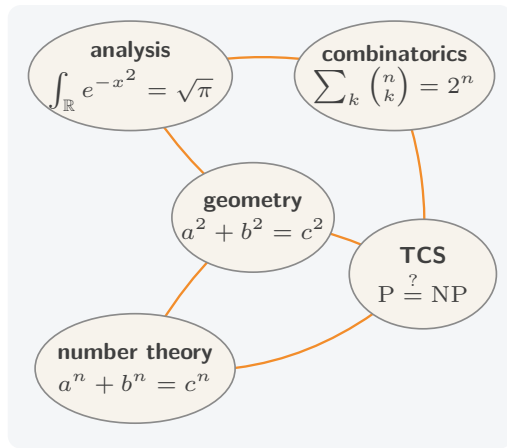
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disproven using an internal reasoning model



AI connects areas of mathematics
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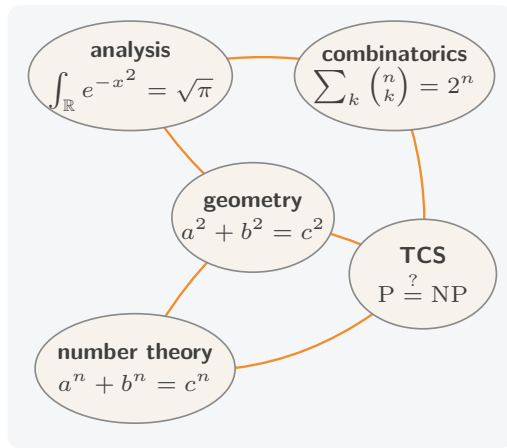
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- ▶ Models know *every* island of mathematics, while most humans know at most a few



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Potential of LLMs

- ▶ Unit-distance conjecture (May 2026): disproven using an internal reasoning model
- ▶ Models know *every* island of mathematics, while most humans know at most a few
- ▶ Cross-island connections: underexplored, and suddenly cheap



AI connects areas of mathematics
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The pinnacle: an asymptotic construction

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- ▶ Clever application of the Central Limit Theorem: projects the infinite-dimensional construction back down to low dimensions
- ▶ 10^{-4} improvement
- ▶ Generalization of Krivine rounding schemes — and a whole new search space

One partition, many shadows

The construction lives in infinitely many dimensions — we only ever see two-dimensional shadows of it.

Colour: which side of the partition dominates. Brightness: how much Gaussian mass lands there.



guided loop



wide tour



double orbit

Three paths through the space of 2D projections — the same infinite-dimensional partition throughout

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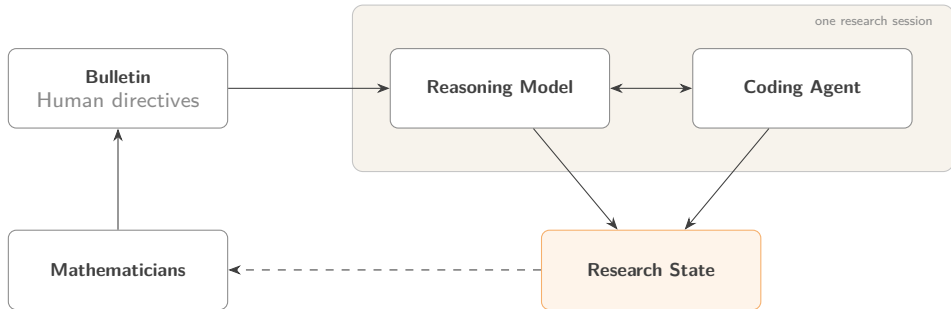
From chat to orchestration

- ▶ AI has too many ideas to explore manually

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- ▶ Needed: persistence, parallelism, orchestration

The harness



- Long-horizon research: iterative agent instantiations with persistent research state

New lower bound on K_G

- ▶ $K_G \geq 6\pi/11 = 1.71359\dots$ (previous best: 1.67696)

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- ▶ Achieved through several human interventions

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- ▶ Represents the current boundary of open-endedness of AI

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- ▶ A fully autonomous AI mathematician?

$$\frac{6\pi}{11} \leq K_G \leq \frac{\pi}{2 \log(1 + \sqrt{2})} - 10^{-4}$$

thank you

New Lower and Upper Bounds for the Grothendieck Constant — arXiv:2608.11158

Long-Horizon AI Research for the Grothendieck Constant — arXiv:2608.11195

$K_G < \pi/(2 \log(1 + \sqrt{2})) - 10^{-5}$ — arXiv:2606.03991



The papers



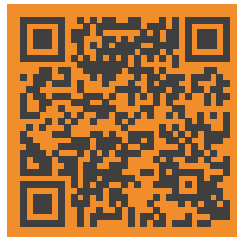
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